

PAWAN KUMAR

Lead Geospatial Analyst, R&D - Geospatial

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PROFESSIONAL SUMMARY

Lead Geospatial Analyst with expertise in automating Geospatial production, QC/QA workflows, and operational processes. Proven track record of developing **25+ Apps/Tools/Plugins & 50+ scripts** to **automate** the traditional manual process and achieving **30-40% reduction in time and costs** for geospatial operations. Specialized in Python-based automation, spatial data engineering, and AI-powered workflow optimization. Proficient in leveraging modern AI tools (ChatGPT, Claude AI, Cursor AI, Kiro) for rapid development and problem-solving.

CORE COMPETENCIES

Geospatial & Remote Sensing: HR Imagery Analysis & Processing • LiDAR Data Processing for Elevation • CRS & Data Transformation • ArcGIS Pro • QGIS • Google Earth Engine • ERDAS Imagine

Programming & Automation: Python (GeoPandas, Rasterio, GDAL, PDAL, Shapely, ArcPy, PyQGIS) • AWS (S3) • Git/GitHub • OSGeo4W • Workflow Automation

AI & Development Tools: ChatGPT • Claude AI • Cursor AI • Kiro AI • GitHub • AI-Assisted Development • Prompting

PROFESSIONAL EXPERIENCE: [Total Exp- 15+ years, Geospatial Exp- 10+ years]

Lead Geospatial Analyst, (R&D – Geospatial), AiDash, [Feb 2022] – Present | Bangalore, India

- Delivered 25+ production-grade geospatial tools to automate repetitive workflows, reducing manual effort by 30–40%.
- Built 50+ Python automation scripts for data processing, QA/QC, and spatial analysis, cutting operational time and costs significantly.
- Developed custom plugins for QGIS and ArcGIS Pro to streamline data labeling, validation, and analysis at scale.
- Created an automated reporting system for real-time monitoring of production metrics and quality indicators.
- Established robust QA/QC frameworks and validation protocols to improve data accuracy and reduce errors.
- Optimized raster and vector workflows using parallel processing and efficient algorithms, boosting speed by up to 50%.
- Acted as the primary technical expert, resolving complex spatial data issues and production bottlenecks.
- Collaborated with engineering, product, and operations teams to integrate geospatial solutions into enterprise systems.

KEY PROJECTS

1. **Intelligent Vegetation Management System (IVMS)** at AiDash:
It uses HR satellite imagery & AI/ML models to detect trees, growth patterns, and proximity to assets. The system predicts future vegetation encroachment risks rather than just current conditions. It helps utilities plan targeted trimming and maintenance, reducing outages and costs. A major benefit is wildfire risk reduction, by identifying dry/high-risk vegetation near lines and enabling proactive clearance. It improves grid reliability, safety (including fire prevention), and operational efficiency.
2. **3D Digital GeoTwin** (Virtualize and Analyse the Real World) at Genesys:
As we experience the uncertainties of climate change, global warming, resources depletion, rapid urbanization, the real-world 3D virtual model as Digital GeoTwin is the answer to these. Merging the physical with the virtual, 3D Digital GeoTwin is going to steer us towards a seamlessly connected real- virtual world, where one can plan, design, model, build, maintain, and sustain environmentally balanced world. Whether it is modelling for floods and inundation in coastal cities or mapping the urban heat, vulnerability hot spots and environmental quality in real time.
3. **Crop Acreage, Yield & Estimation** at RMSI:
It estimates farm-level yield, remotely identifies loanee acreage on map and tracks crop health (analyses portfolio risk) as the season progresses, estimates loan-wise loss over hundreds of thousands of crop loans

remotely. Draws a list of loanee farmers who are likely to be distressed and more importantly, unlikely to receive claim payouts and identifies faulty crop loan applications.

4. **GIM (Geo Intelligence Mission)** at DRDO – Stesalit:

The GIM will enable the automatic extraction of data on man-made and natural terrain features, including roads, habitations, forests, hills, and rivers. Flood mapping for strategic perception and geo-visualization will also be done. The Defence Terrain Research Laboratory focuses on providing state-of-the-art terrain intelligence to the armed forces. With knowledge of the terrain. A lot of deception can take place with the help of terrain. The mission is to become technologically self-reliant in using high resolution terrain-intelligence products. This will involve creating and updating thematic maps and terrain intelligence reports for the services.

5. **Study of drinking water quality analysis** (Korba block) at Ceinsys:

Korba district is known for coal mines and power generation plants. So, it resulted in groundwater pollution and surface water contamination. We studied the background of water pollution and conceptualized the project idea for water quality mapping.

6. **Financial inclusion** in rural India at Vakrangee: India has focused on exploring cost-effective and sustainable delivery mechanisms to reach out to remote locations. Financial inclusion is the key to bridging the social divide and achieving well-distributed, robust, and sustainable economic growth.

7. **UIDAI(AADHAR)** at Lakshya Infrapower: The UIDAI program has been to provide a universal identity to every Indian resident. The card alone will suffice to serve as proof of identity, and proof of address. It is expected to help reduce corruption since every individual will have only one unique number. Aadhaar may be given more prominence in the years to come as more government schemes are being launched that require it as a mandatory document.

EDUCATION

M. Tech. - Master of Technology in **Remote Sensing & GIS**, 2011 | Coimbatore

DRDO BU CLS Coimbatore, TN, India

Certificate course in **Climate Change: Carbon Capture & Storage**, 2024 | Online

The University of Edinburgh, UK

M. Sc. – Master of Science in **Chemistry** 2008 | Kanpur

CSJM University, Kanpur, UP, India

B. Sc. – Bachelor of Science in **PCM** 2006 | Auraiya

CSJM University, Kanpur, UP, India

CERTIFICATIONS

- AI - Summer Training Program 2025 (Hybrid) – **Indian Institute of Technology, Kanpur** (Aug 2025)
- Python for Data Science – **Indian Institute of Technology, Kanpur** (Aug 2024)
- Essence of Deep Learning & Recent Trends (**EDRT-2024**) – **Annant Gyan Knowledge and Skills Pvt Ltd**
- Ecological Risk Management & Sustainable Earth – **IQAC Tilak Mahavidyalaya Auraiya** (Mar 2024)
- Fundamentals of Machine Learning for Earth Science – **NASA** (Apr 2023)
- Human Values & Professional Ethics – **Annant Gyan Knowledge and Skills Private Limited** (Jul 2023)
- End-to-End Google Earth Engine – **Spatial Thoughts** (Apr 2023)
- Modern Instrumental & Characterization Techniques in Applied Sciences (**MICTAS-2020**) – **MIET**
- Excel (Beginner to Advanced:Data Analytics & Visualization) **Annant Gyan Knowledge and Skills P Ltd** (2022)
- Machine Learning Pipelines with Azure ML Studio – **Coursera** (Oct 2022)
- Python Program from UST – **Coursera** (Oct 2022)
- Machine Learning to Deep Learning for Remote Sensing – **ISRO** (Jul 2022)
- Hands-on Python & Machine Learning – **Annant Gyan Knowledge and Skills Private Limited** (Feb 2022)
- STC on Remote Sensing & Applications – **Lovely Professional University** (Feb 2022)
- Workshop on Atmospheric Sciences – **DRDO-ISRO** (Jun 2010)

ADDITIONAL INFORMATION

Won 2nd prize in the category "**Best Operations Game Changer Hacks**" at **Aidash's** first-ever internal '**Hackathon-25**' for automation in current QA process in GIS.

Languages: English (Fluent), Hindi (Native)
